FORM FOUR
INFORMATION AND COMPUTER STUDIES (ICS)
EXCEL PRACTICAL EXAMINATION
PRE-NECTA PRACTICE PAPER

Time: 2 Hours	Maximum Marks: 50	Paper: Practical
Candidate Name: _____	Class: Form Four	Date: _____

INSTRUCTIONS TO CANDIDATES

1. This is a practical examination. Perform all tasks using Microsoft Excel or compatible spreadsheet software.
2. Create and save your workbook as **FORM4_ICS_EXCEL** in the location instructed by the teacher.
3. Use the data supplied in Question 1. Do not add unnecessary records.
4. Enter formulas where calculations are required. Do not type calculated answers manually.
5. Unless otherwise stated, use the appropriate cell references so that formulas can be copied down or across.
6. Save your work regularly and check all results before submitting.
7. This paper is designed as a Form Four **pre-NECTA practice examination**; it is not an official NECTA examination paper.

QUESTION 1: DATA ENTRY AND WORKSHEET PREPARATION (10 marks)

Open a new workbook and rename Sheet1 as **STUDENT SALES**. Enter the following data exactly as shown.

No.	Student	Gender	Item	Units	Selling Price	Cost Price	Marks
1	Asha	F	Exercise Book	8	2500	1800	78
2	Baraka	M	Pen	12	1200	700	64
3	Cynthia	F	Ruler	10	1500	900	52
4	David	M	Exercise Book	6	2500	1800	45
5	Esther	F	Pen	15	1200	700	81
6	Faraja	M	Ruler	7	1500	900	39
7	Grace	F	Exercise Book	9	2500	1800	67
8	Hassan	M	Pen	11	1200	700	58
9	Irene	F	Ruler	5	1500	900	72
10	Juma	M	Exercise Book	4	2500	1800	49

(a) Apply a suitable heading style to the table. Make the headings bold, centred and clearly visible. (2 marks)

(b) Format Selling Price and Cost Price as currency/number with a thousands separator. Adjust column widths and row heights so all information is visible. (2 marks)

(c) Apply borders to the whole table and use an appropriate alignment for text and numerical data. (2 marks)

(d) Add a suitable title above the table: **STUDENT SALES AND PERFORMANCE RECORD**. Merge and centre the title across the table. (2 marks)

(e) Save the workbook. (2 marks)

QUESTION 2: FORMULAS, REFERENCES AND CONDITIONAL LOGIC (15 marks)

Add the following columns after Marks: **Total Sales, Total Cost, Profit/Loss, Profit %, Grade, Result**.

(a) Calculate **Total Sales** using Units × Selling Price. (2 marks)

(b) Calculate **Total Cost** using Units × Cost Price. (2 marks)

(c) Calculate **Profit/Loss**. A positive result represents profit and a negative result represents loss. (2 marks)

(d) Calculate **Profit %** as Profit/Loss divided by Total Cost × 100. Round the answer to 2 decimal places. (2 marks)

(e) Use an **IF** formula to display **PASS** when Marks are 50 or above and **FAIL** when Marks are below 50. (2 marks)

(f) Use a suitable formula to assign grades: **A** for 75–100, **B** for 65–74, **C** for 50–64, and **F** for below 50. (3 marks)

(g) Copy the formulas down for all ten students. Use cell references correctly. (2 marks)

QUESTION 3: LOOKUP, COUNTING AND CONDITIONAL SUMMING (10 marks)

Create a small lookup table on the same worksheet or on a new sheet named **LOOKUP**.

Grade	Description	Point
A	Excellent	4
B	Very Good	3
C	Pass	2
F	Fail	0

(a) Use **VLOOKUP** to return the Description for each student's Grade. (3 marks)

(b) Use **COUNTIF** to determine how many students obtained each grade A, B, C and F. (2 marks)

(c) Use **COUNTIF** to determine the number of students who passed. (1 mark)

(d) Use **SUMIF** to calculate the total Units sold for **Exercise Book**. (2 marks)

(e) Use **AVERAGE**, **MAX** and **MIN** to find the average, highest and lowest Marks. (2 marks)

QUESTION 4: DATA VALIDATION AND CONDITIONAL FORMATTING (8 marks)

(a) Create a **drop-down list** for Gender containing only **M** and **F**. (2 marks)

(b) Apply **Data Validation** to Marks so that only whole numbers from **0 to 100** can be entered. Add an appropriate input message and error alert. (2 marks)

(c) Apply **Conditional Formatting** to Marks to clearly identify students who have failed (below 50). (2 marks)

(d) Apply another conditional formatting rule to identify the highest mark in the list. (2 marks)

QUESTION 5: SORTING, FILTERING AND CHART (7 marks)

(a) Sort the complete table by Marks from **Largest to Smallest**, keeping each student's information together. (2 marks)

(b) Use Filter to display only students who **PASS**. (1 mark)

(c) Remove the filter and filter the Item column to display only **Pen** records. (1 mark)

(d) Create a suitable **column chart** showing Student names against Marks. Include a chart title, axis titles and data labels where appropriate. (3 marks)

QUESTION 6: RELATIVE, ABSOLUTE AND MIXED REFERENCES (BONUS/CHALLENGE – 5 marks)

In a separate area of the worksheet, enter **VAT Rate = 18%** in one cell. Using formulas, calculate VAT for each student's Total Sales. Use an **absolute cell reference** for the VAT rate so the same formula can be copied down. Then calculate Sales Including VAT.

(a) Demonstrate the correct use of an absolute reference for the VAT rate. (2 marks)

(b) Calculate Sales Including VAT using the Total Sales and VAT amount. (2 marks)

(c) Demonstrate that the formula can be copied down without changing the VAT-rate reference. (1 mark)

FINAL CHECK BEFORE SUBMISSION

Check	Candidate tick
All required data entered correctly	■
Formulas used instead of manually typing calculated answers	■
Data validation and conditional formatting applied	■
Sorting/filtering completed correctly	■
Chart created and labelled	■
Workbook saved as FORM4_ICS_EXCEL	■

END OF EXAMINATION

Teacher's remark: _____

Candidate's signature: _____ Teacher's signature: _____